Supplementary Information

Computational Design of 106 Novel, Brain-Penetrant Acetylcholinesterase Inhibitor Candidates with Computed Synthetic Routes

U. Raj, M. Aggarwal, I. Sharma, C. M. Dasari

S1. Pose-stability dynamics

Pose stability (Supplementary Table 1). Donepezil from its crystal pose stayed in the gorge (mean RMSD over the second half 2.12 Å), so the protocol is valid for this replica. Of the 17 shortlisted molecules simulated, 16 kept their pose by the criterion and 1 did not (Shortlist 12, RMSD 4.16 Å). Shortlist 18, 19 and 20 were not simulated in this first replica because its time budget was used; the second replica below covers them. One 2 ns replica is a short and single test (a second replica follows), so this is a screen for gross pose instability and not a free-energy or residence-time result; no molecule's potency can be inferred from it.

Supplementary Table 1. Pose stability in 2 ns of explicit-solvent dynamics (first replica, seed 1). RMSD: mean ligand heavy-atom RMSD over the second half of production (Å); COM: centre-of-mass shift over the same period (Å); Trp86: fraction of frames with a ligand aromatic ring centroid within 5.5 Å of the Trp86 ring centroid (all values here were recovered from the saved trajectories and the run log, because the summary file of one of the two GPU processes was overwritten by the other; the remaining contact fractions and the MM-GBSA energies survive for only half of the molecules and are therefore not reported). Stable: RMSD at most 3.0 Å and centre-of-mass shift at most 6 Å.

Molecule	RMSD	COM	Trp86	Stable
donepezil (crystal pose)	2.12	0.56	0.96	yes
Shortlist 1	1.08	0.70	0.99	yes
Shortlist 2	2.64	2.04	0.98	yes
Shortlist 3	0.99	0.79	0.00	yes
Shortlist 4	2.53	0.70	0.78	yes
Shortlist 5	1.23	0.67	1.00	yes
Shortlist 6	1.48	1.01	0.00	yes
Shortlist 7	2.17	1.39	0.00	yes
Shortlist 8	0.99	0.64	0.99	yes
Shortlist 9	2.41	2.15	0.00	yes
Shortlist 10	0.71	0.41	0.00	yes
Shortlist 11	1.47	0.83	0.00	yes
Shortlist 12	4.16	2.95	0.00	no
Shortlist 13	1.98	1.57	0.00	yes
Shortlist 14	1.42	0.72	0.00	yes
Shortlist 15	2.27	1.52	0.00	yes
Shortlist 16	1.33	0.99	1.00	yes
Shortlist 17	0.99	0.62	0.00	yes

Replicate. A second 2 ns replica (seed 2, identical poses and protocol, no MM-GBSA) covered all 20 shortlisted molecules, including Shortlist 18 to 20, which the first replica did not reach (Supplementary

Table 2). In the replica donepezil, the reference that must pass for the protocol to be valid, had a mean RMSD of 3.13 Å (2.12 Å in the first replica), just over the 3.0 Å criterion, so the protocol did not meet its own validity condition in this replica. The difference between the two donepezil trajectories (about 1 Å) is the run-to-run variation of a single 2 ns trajectory, and every stable or unstable call below carries it. In the replica 19 of 20 shortlisted molecules met the stability criterion. For the 17 molecules simulated twice, 15 were stable in both replicas and the call agreed for 15; the calls differed for Shortlist 12 (RMSD 4.16 and 1.28 Å) and Shortlist 13 (RMSD 1.98 and 3.81 Å). The Spearman correlation between the RMSD values of the two replicas is only 0.11 (mean absolute difference 0.90 Å). The dynamics therefore detect gross instability reproducibly, since no shortlisted molecule left the gorge in either replica (largest centre-of-mass shift 3.3 Å), but a 3.0 Å RMSD criterion on a 2 ns trajectory does not separate individual molecules reliably, and the first-replica statement about Shortlist 12 should be read with this variation. Two replicas of 2 ns remain a screen, and agreement between them would say only that the screen is repeatable, not that the poses are correct.

Supplementary Table 2. Replicate pose-stability dynamics (2 ns each; seeds 1 and 2). RMSD: mean ligand heavy-atom RMSD over the second half of production (Å); Stable: RMSD at most 3.0 Å and centre-of-mass shift at most 6 Å. The first replica did not reach Shortlist 18 to 20 (–).

Molecule	RMSD 1	RMSD 2	Stable 1	Stable 2
donepezil (crystal pose)	2.12	3.13	yes	no
Shortlist 1	1.08	0.75	yes	yes
Shortlist 2	2.64	2.70	yes	yes
Shortlist 3	0.99	1.98	yes	yes
Shortlist 4	2.53	1.59	yes	yes
Shortlist 5	1.23	2.15	yes	yes
Shortlist 6	1.48	2.43	yes	yes
Shortlist 7	2.17	1.75	yes	yes
Shortlist 8	0.99	1.58	yes	yes
Shortlist 9	2.41	1.92	yes	yes
Shortlist 10	0.71	2.29	yes	yes
Shortlist 11	1.47	2.52	yes	yes
Shortlist 12	4.16	1.28	no	yes
Shortlist 13	1.98	3.81	yes	no
Shortlist 14	1.42	1.15	yes	yes
Shortlist 15	2.27	1.01	yes	yes
Shortlist 16	1.33	1.89	yes	yes
Shortlist 17	0.99	0.83	yes	yes
Shortlist 18	–	0.84	–	yes
Shortlist 19	–	0.65	–	yes
Shortlist 20	–	1.17	–	yes

Validation of the screen against known activity. To test whether the pose-stability screen carries any information about activity, we applied the identical protocol (docking, preparation, 2 ns explicit-solvent run, one trajectory each) to ten potent and ten inactive published AChE compounds (Supplementary Table 3). The screen did not separate them. The criterion RMSD at most 3.0 Å with centre-of-mass shift at most 6 Å was met by 10 of 10 potent and 8 of 10 inactive compounds (Fisher test, $p = 0.47$); the

second-half RMSD gave an area under the curve of 0.47 (95% interval 0.20 to 0.74, Mann–Whitney p = 0.85) and the centre-of-mass shift 0.63 (0.38 to 0.87, p = 0.34), with the potent compounds having the slightly higher median RMSD (1.52 against 1.33 Å). Donepezil in its crystal pose again exceeded the RMSD criterion in this run (3.29 Å), so the protocol did not meet its own validity condition here either. Persistent stacking on Trp86 was the one measure that differed (potent compounds, median fraction 0.99; inactive compounds, 0.35; area under the curve 0.81, 95% interval 0.58 to 0.99, p = 0.018); it was one of three measures examined, the test is not corrected for that, and it rests on ten compounds per class, so it is a lead for further work and not a demonstrated discriminator. We therefore state that the stability screen, as run, is a check against gross loss of the pose and not a predictor of potency, and no ranking in this paper depends on it.

Supplementary Table 3. Does the pose-stability screen separate potent from inactive published AChE compounds? Ten potent (pChEMBL at least 8.5) and ten inactive (below 5) compounds from ChEMBL were docked and simulated with the protocol of the shortlist (one 2 ns trajectory each, seed 7). RMSD and COM shift: second half of production (Å); Trp86: fraction of frames with a ligand aromatic ring centroid within 5.5 Å of the Trp86 ring centroid.

Compound	Class	pChEMBL	RMSD	COM	Trp86
1	potent	9.1	2.48	1.47	0.99
2	potent	8.9	1.03	0.46	0.99
3	potent	9.1	1.30	0.79	0.99
4	potent	8.5	2.34	0.72	0.99
5	potent	8.6	1.17	0.62	1.00
6	potent	8.9	1.45	0.52	0.96
7	potent	9.0	2.08	1.23	1.00
8	potent	9.1	1.90	0.62	1.00
9	potent	8.6	1.58	0.68	0.99
10	potent	9.5	1.18	0.52	1.00
11	inactive	4.7	3.16	1.43	0.00
12	inactive	4.9	2.36	1.32	0.00
13	inactive	4.8	1.15	0.66	0.69
14	inactive	4.5	1.40	0.57	1.00
15	inactive	4.5	3.79	1.96	1.00
16	inactive	4.9	1.04	0.70	0.00
17	inactive	4.5	1.17	0.61	0.99
18	inactive	4.4	2.26	1.25	0.98
19	inactive	4.5	1.26	0.94	0.00
20	inactive	4.6	0.69	0.47	0.00

S2. Anchored shortlist

Supplementary Table 4. The shortlist anchored to measured AChE data. Parent: measured pChEMBL of the potent published parent. Edits: the one or two matched-pair edits. Nearest measured: ECFP4 similarity and pChEMBL of the most similar measured AChE compound. P(active): activity-model probability.

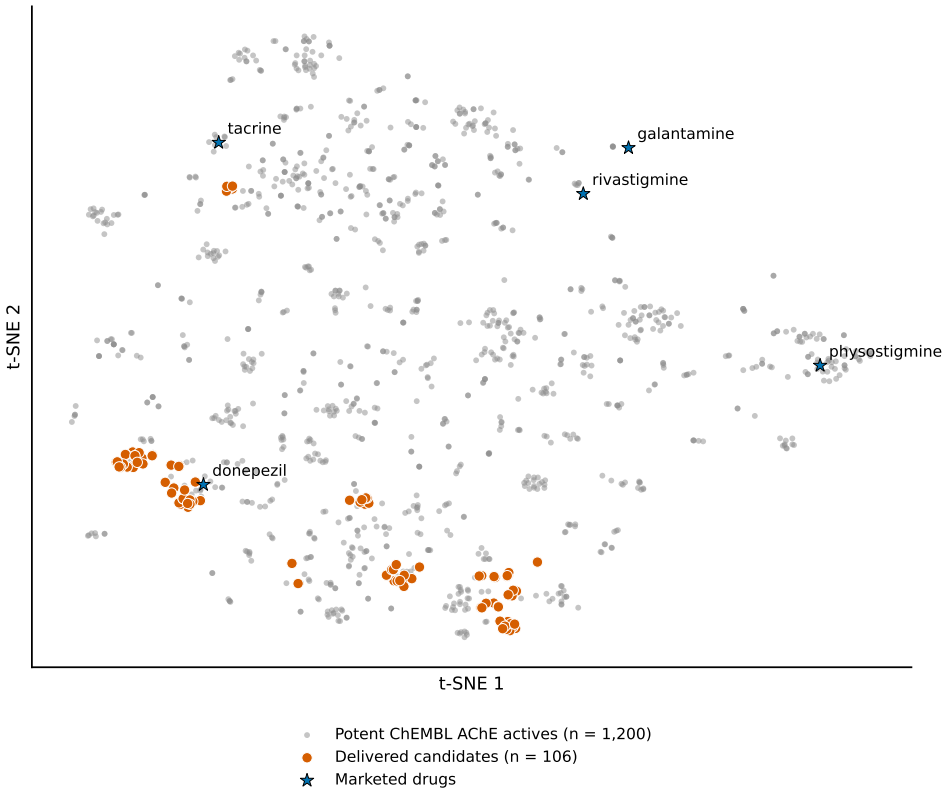
Rank	Parent	Edits	Nearest measured	P(active)
1	6.8	OMe→SMe	0.83 / 6.8	0.90
2	6.8	OMe→SMe	0.83 / 6.8	0.90
3	8.4	Ph→Me, Me→cPr	0.73 / 8.4	0.90
4	6.8	Me→cPr	0.90 / 6.8	0.91
5	6.9	Me→CH2Cl, CH2Cl→CH2OH	0.83 / 6.9	0.94
6	7.8	Ph→Me, Me→CH2cPr	0.74 / 7.8	0.89
7	7.8	Ph→Me, Me→cPr	0.74 / 7.8	0.90
8	6.8	Me→cPr	0.90 / 6.8	0.91
9	7.6	Me→cPr, Ph→Me	0.67 / 8.1	0.90
10	–	Me→cPr, Ph→Me	0.48 / 8.1	0.89
11	6.9	OMe→Me	0.86 / 6.9	0.86
12	8.4	Ph→Me	0.73 / 8.4	0.84
13	7.6	Me→cPr, Ph→Me	0.67 / 8.1	0.90
14	8.4	Ph→Me, Me→CH2cPr	0.73 / 8.4	0.89
15	–	Me→cPr, Ph→Me	0.48 / 8.1	0.89
16	7.5	Ph→2-F-Ph	0.75 / 7.5	0.94
17	8.4	Ph→Me, Et→Pr	0.71 / 8.4	0.88
18	6.6	OMe→Me	0.87 / 6.6	0.86
19	6.7	Ph→Me, Et→Pr	0.74 / 6.7	0.77
20	7.8	Ph→Me	0.74 / 7.8	0.86

S3. Blind re-discovery of later-reported compounds

Blind re-discovery. Using only compounds reported up to a cutoff year, we applied the design stage's edits to the potent parents and asked how many compounds reported later were reproduced exactly and how close the generated analogues came to later potent compounds: at the 2015 cutoff the design-stage edits generated 325,244 analogues from 1,247 earlier potent parents and reproduced 7 of the 2,784 later compounds exactly (5 of the 593 later potent compounds), against 5 to 8 (2 to 3 potent) for 3 random draws of unused edits; the share of later potent compounds with a generated analogue at ECFP4 similarity of at least 0.8 was 2.9 % for the design-stage edits and 2.0 to 2.9 % for the random draws (mean highest similarity 0.48 against 0.47); at the 2018 cutoff the design-stage edits generated 345,702 analogues from 1,531 earlier potent parents and reproduced 6 of the 1,819 later compounds exactly (3 of the 393 later potent compounds), against 4 (1 to 3 potent) for 3 random draws of unused edits; the share of later potent compounds with a generated analogue at ECFP4 similarity of at least 0.8 was 2.5 % for the design-stage edits and 1.5 to 2.3 % for the random draws (mean highest similarity 0.49 against 0.47 to 0.48); at the 2012 cutoff (exact matches only) the design-stage edits reproduced 5 later compounds (4 potent) against 3 to 8 (2 to 3 potent) for 2 random draws. Exact re-discovery is rare and the counts are too small to separate the edit sets, and the near-recovery measure shows at most a small difference (the design-stage

edits are within the range of the random draws at 2015 and slightly above it at 2018): the edit library has not been shown to find future compounds clearly better than other edits, which agrees with the matched-pair statistics test of Table 1 of the main paper. The value of the cascade therefore lies in its gates and its anchoring to measured parents and not in the edit selection.

S4. Chemical space of the delivered candidates



Supplementary Figure 1. Chemical space of the delivered candidates. t-SNE embedding (Jaccard distance on ECFP4) of 1,200 potent ChEMBL AChE actives, the delivered candidates and five marketed drugs.